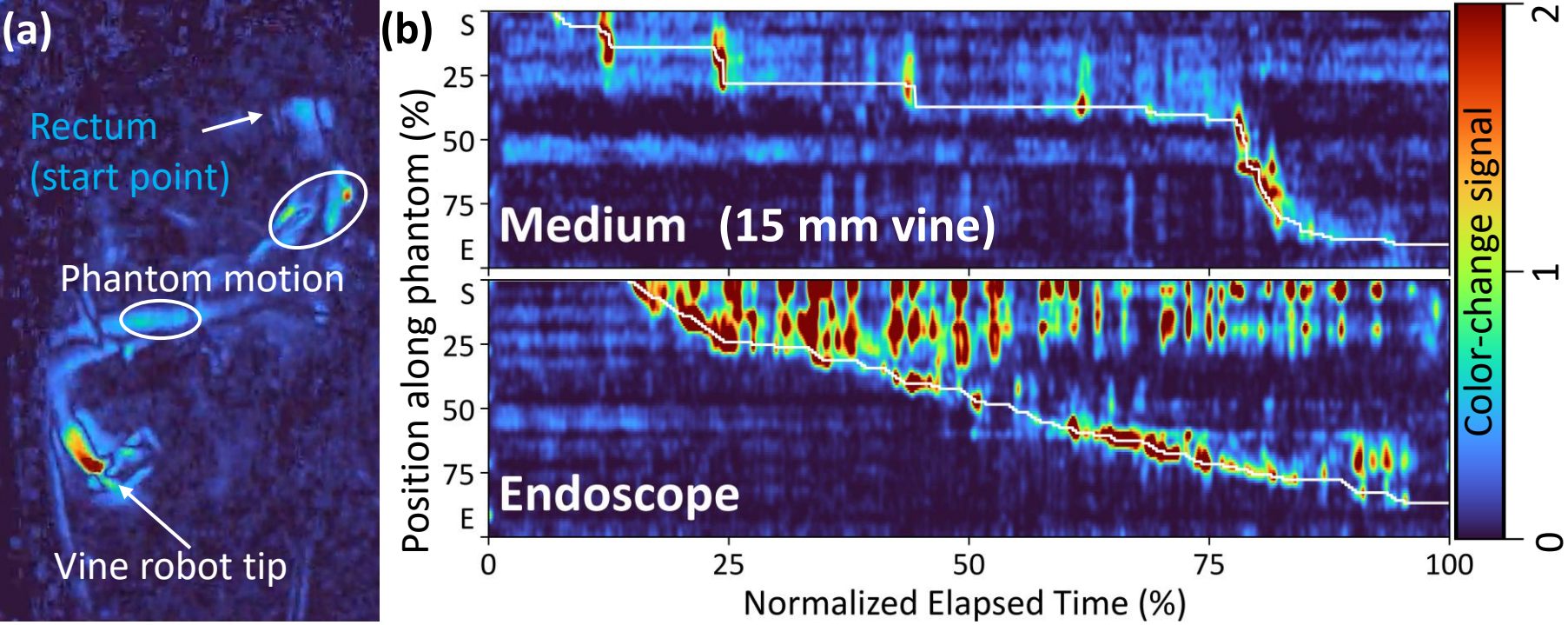

(a)
Rectum
(start point)
Phantom motion
Vine robot tip
(b)
S
25
50
75
E
Medium (15 mm vine)
Endoscope
Position along phantom (%)
0
25
50
75
100
Normalized Elapsed Time (%)
Color-change signal
0
1
2